

DATA INPUTS

SPECIES DISTRIBUTIONS

Range maps for 1,192 species

Global IUCN range maps projected onto a 0.5° ocean grid, forming the spatial foundation for all analyses.

PROTECTED AREAS

Marine Protected Areas

Global database of no-take MPAs, areas where all extractive activities are prohibited.

CONSERVATION INDICES

Three complementary indices

Extinction risk

IUCN Red List status. How threatened is each species?

Ecological role

FUSE. How unique and specialised is each species' function, weighted by extinction risk?

Evolutionary heritage

EDGE2. How ancient and distinct is each species' lineage, weighted by extinction risk?

HUMAN FOOTPRINT

Global fishing pressure

Vessel tracking (AIS) and satellite radar (SAR) combined to estimate industrial fishing pressure across all ocean regions.

ANALYSES

GAP ANALYSIS

Where are the protection gaps?

We calculate how much of each species' range falls within no-take MPAs, and compare this to what would be expected by chance.



CONSERVATION PRIORITISATION

Where should we protect?

Using the CAPTAIN reinforcement learning algorithm, we identify which ocean areas would maximise conservation benefit, separately for each of the three indices, within a realistic 10% protection budget.



CONFLICTS AND OPPORTUNITIES

Where are the conflicts and opportunities?

We overlay conservation priority maps with fishing pressure to identify areas where protection is urgent but difficult, and areas where it would be feasible with minimal disruption.

OUTPUTS

OUTPUT 1

A global picture of protection shortfalls

Species-level and region-level assessment of how poorly current MPAs serve elasmobranch conservation.

OUTPUT 2

Priority maps across three indices

Spatial maps identifying where ocean protection would deliver the greatest benefit for threatened, functionally unique and specialised, and evolutionarily distinct species, and where all three converge.

OUTPUT 3

An evidence base for strategic action

A clear distinction between areas ripe for proactive protection and areas requiring intensive management, informing where and how conservation resources should be deployed.